

12 (**long**) Figure 8 shows an electrical circuit comprising an ideal current source i_s in combination with a capacitor C in parallel with a series combination of a resistor R and inductor L .

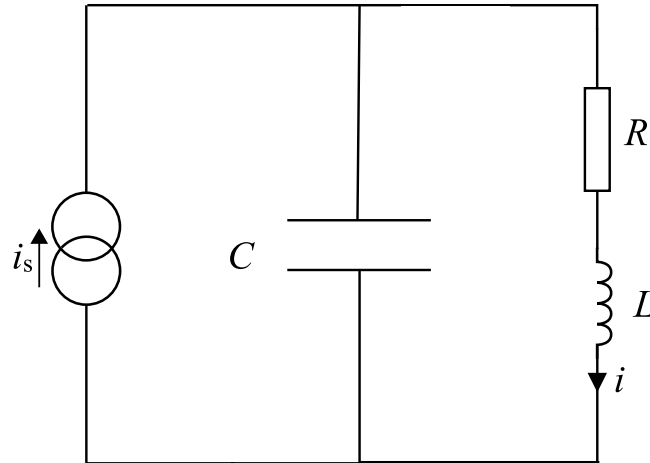


Fig. 8

- (a) Show that an equation for the inductor current i can be derived as [5]

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = \frac{1}{LC} i_s.$$

- (b) Obtain expressions for the natural frequency and damping factor for the system assuming $\frac{1}{LC} = 10^{12} \text{ s}^{-2}$ and $\frac{R}{L} = 10^6 \text{ s}^{-1}$. [5]

- (c) Obtain an expression for the steady-state voltage across the resistor if the current source is given by $i_s = i_0 \sin \omega t$. Estimate a value for the maximum voltage across the resistor for all input frequencies ω if R is 100Ω and i_0 is 10 mA . [10]

- (d) Using the databook or otherwise, estimate the maximum value of the voltage across the resistor if the input source is a unit step function. [5]

- (e) What value would R have to be increased to in order to limit the maximum voltage excursion in (d) to 10% of the nominal final value? [5]

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Numerical Answers:

4(a) 82.9 kJ kg^{-1}

(b) $T_2=222\text{K}$, $q=41.4 \text{ kJ kg}^{-1}$

5(b) 46W

6 (b)(i) 53.6 kg s^{-1}

(ii) 460.3 ms^{-1}

(iii) $-19.1 \text{ J kg}^{-1} \text{ K}^{-1}$

(iv) -0.732 kW

7(a) 1.12R

(b) 5.67V

10(f)(ii) 0.2s

12) (b) 10^6rad s^{-1} , 0.5

c) 1.16V

d) 117V

(e) 120Ω